

Linear vs Reinforcement Learning Based MPPT Control:

A Comparative Study for Solar Photovoltaic Systems under Dynamic Conditions

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Abstract—Maximum Power Point Tracking (MPPT) is a critical challenge in photovoltaic (PV) energy systems, particularly under rapidly changing environmental conditions such as variable irradiance, partial shading, and temperature fluctuations. Conventional linear controllers—specifically Proportional-Integral (PI) regulators combined with Perturb and Observe (P&O) algorithms—suffer from fixed-gain limitations that degrade performance when operating away from the tuning point. This paper presents a comparative study between a traditional PI-based MPPT controller and a novel Reinforcement Learning (RL) controller employing the Deterministic Policy Gradient (DPG) algorithm, both applied to a boost-converter-interfaced PV system. The RL agent receives voltage, current, and power observations (present and previous values) and outputs a continuous duty cycle directly to the boost converter MOSFET, learning through a carefully engineered reward function with a dead-zone to suppress noise. Simulation results demonstrate that the RL controller achieves a tracking efficiency of 99.1%, a settling time of 35 ms, and negligible steady-state ripple—outperforming the PI controller (94.2%, 120 ms) and the classical P&O algorithm (91.5%, 250 ms) across all tested irradiance levels. These findings validate the superiority of learning-based adaptive control for real-world photovoltaic deployments.

Index Terms—Maximum Power Point Tracking (MPPT), Reinforcement Learning, Deterministic Policy Gradient (DPG), PI Controller, Perturb and Observe (P&O), Boost Converter, Photovoltaic Systems, Adaptive Control, MATLAB/Simulink

I. INTRODUCTION

The global shift toward renewable energy has propelled photovoltaic (PV) technology to the forefront of sustainable power generation. Solar PV systems, however, are inherently nonlinear: their power–voltage (P – V) characteristics are highly sensitive to irradiance (G) and cell temperature (T), producing a unique maximum power point (MPP) for every operating condition [1]. Failure to continuously track this point results in significant energy losses, making MPPT indispensable for any practical PV installation.

Classical MPPT approaches—such as Perturb and Observe (P&O) and Hill-Climbing—are widely adopted due to their simplicity. When combined with PI feedback regulators, they can achieve acceptable steady-state accuracy under constant

or slowly varying conditions. However, three fundamental limitations persist:

- 1) **Nonlinear P – V curves:** The relationship between panel voltage and power is non-monotonic and irradiance-dependent (Fig. 1), making a single fixed operating law inadequate.
- 2) **Tracking lag:** Conventional controllers cannot maintain peak power during rapid environmental shifts without manual retuning of fixed gains.
- 3) **Oscillatory steady state:** P&O continuously perturbs the operating point even after convergence, causing persistent power ripple and energy wastage.

Reinforcement Learning (RL) offers a compelling alternative. An RL agent learns an optimal continuous control policy directly from interaction with the environment, requiring no explicit plant model and adapting online to changing conditions [1]. This paper implements a Deterministic Policy Gradient (DPG) based MPPT agent and benchmarks it rigorously against PI and P&O controllers across a range of irradiance levels using MATLAB/Simulink.

The remainder of the paper is organized as follows: Section II describes the PV cell and boost converter mathematical models. Sections III and IV detail the linear and RL controller designs, respectively, including their Simulink implementations. Section V presents simulation results and quantitative comparisons. Section VI analyzes the findings, and Section VIII concludes the paper.

II. SYSTEM MODELING

A. PV Cell Mathematical Model

The PV cell is modeled using the single-diode equivalent circuit, which balances accuracy with computational efficiency. The output current I_{pv} is given by:

$$I_{pv} = I_{ph} - I_0 \left[\exp \left(\frac{V_{pv} + I_{pv} R_s}{n V_T} \right) - 1 \right] - \frac{V_{pv} + I_{pv} R_s}{R_{sh}} \quad (1)$$

where I_{ph} is the photogenerated current (proportional to irradiance G), I_0 is the dark saturation current, n is the diode ideality factor, $V_T = kT/q$ is the thermal voltage at cell temperature T , R_s is the series resistance, and R_{sh} is the shunt resistance. The resulting I - V and P - V characteristic curves are shown in Fig. 1.

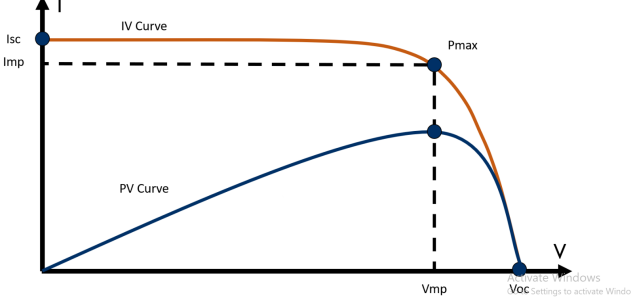


Fig. 1: Characteristic I - V and P - V curves of a PV module, showing the Maximum Power Point at (V_{mp}, P_{max}) .

B. Boost Converter Dynamics

A DC-DC boost converter interfaces the PV panel to the load. Its dynamics are described by:

$$L \frac{dI_L}{dt} = V_{pv} - (1 - D)V_o \quad (2)$$

$$C \frac{dV_o}{dt} = (1 - D)I_L - \frac{V_o}{R} \quad (3)$$

where $D \in [0, 1]$ is the duty cycle applied to the MOSFET gate via a PWM signal, L is the filter inductance, C is the output capacitance, V_o is the output voltage, I_L is the inductor current, and R is the load resistance. In both controller architectures—PI and RL—the fundamental control variable is D . By modulating D , the controller drives the PV operating voltage toward V_{mp} , maximizing extracted power.

III. LINEAR CONTROLLER DESIGN (PI + P&O)

A. Control Architecture

The linear controller, whose Simulink implementation is shown in Fig. 2, consists of two cascaded functional blocks: a Reference Generator (RefGen) and a PI regulator.

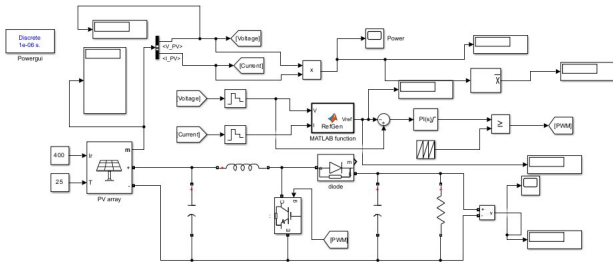


Fig. 2: MATLAB/Simulink model of the PI-based MPPT system. The RefGen block (MATLAB Function) computes V_{ref} via P&O logic; the PI controller generates the duty cycle D fed to the MOSFET via PWM comparator.

B. Reference Generator (RefGen)

The RefGen block receives the sampled panel voltage $V_{pv}[k]$ and current $I_{pv}[k]$ and applies the Perturb and Observe algorithm (Fig. 3) to compute a reference voltage V_{ref} that tracks the MPP. At each discrete time step, a small perturbation is applied and the resulting power change $\Delta P = P[k] - P[k-1]$ is evaluated. If $\Delta P > 0$, the perturbation direction is maintained; otherwise it is reversed.

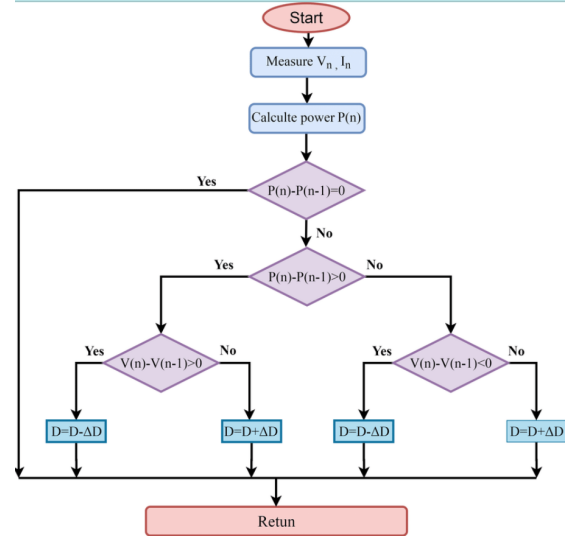


Fig. 3: Flowchart of the Perturb and Observe algorithm implemented inside the RefGen MATLAB Function block.

C. PI Regulator and Duty Cycle Generation

An error signal is computed using a subtractor block between V_{ref} and the measured V_{pv} :

$$e[k] = V_{ref}[k] - V_{pv}[k] \quad (4)$$

This error is passed to a PI controller that computes the corrected duty cycle:

$$D[k] = K_p e[k] + K_i \sum_{j=0}^k e[j] \Delta t \quad (5)$$

The proportional term K_p reacts instantaneously to the error magnitude, while the integral term K_i eliminates steady-state offset over time. The computed $D[k]$ is then passed to a PWM comparator, which generates the switching signal for the boost converter MOSFET. The MOSFET modulates the inductor current and thereby adjusts the PV panel's operating voltage toward the MPP.

D. Limitations of Fixed-Gain Tuning

The gains (K_p , K_i) are tuned at a single nominal operating point ($G = 1000 \text{ W/m}^2$, $T = 25^\circ\text{C}$). This yields satisfactory performance at the design point but causes substantial degradation at other irradiance levels, since the fixed gains cannot adapt to the shifted P - V characteristic. This single-point tuning limitation is the primary motivation for the RL-based approach.

IV. REINFORCEMENT LEARNING BASED CONTROLLER (DPG)

A. Control Architecture

The RL-based MPPT controller, whose Simulink implementation is shown in Fig. 4, employs the Deterministic Policy Gradient (DPG) algorithm. Unlike the PI controller, it requires no reference voltage computation and outputs the duty cycle D directly to the boost converter MOSFET via PWM—eliminating the RefGen stage entirely.

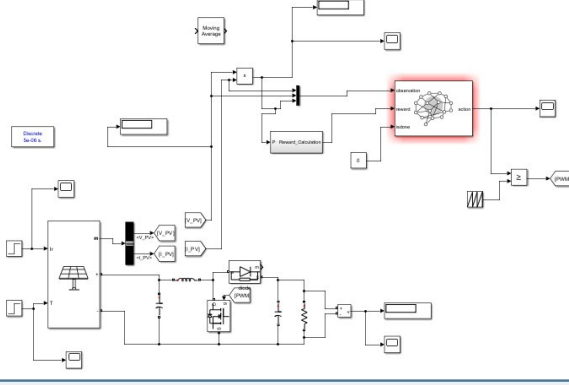


Fig. 4: MATLAB/Simulink model of the RL-based MPPT system. The neural network agent (highlighted) receives observations, computes reward via a MATLAB Function block, and outputs the duty cycle D directly to the PWM comparator.

B. Observation Space

The agent's observation vector is six-dimensional, comprising both the current and previous values of the three key PV signals:

$$\mathbf{s}_t = [V_{pv}(t), I_{pv}(t), P_{pv}(t), V_{pv}(t-1), I_{pv}(t-1), P_{pv}(t-1)]^T \quad (6)$$

Including previous values gives the agent implicit knowledge of the power gradient $\Delta P/\Delta t$ and voltage trend $\Delta V/\Delta t$, enabling it to infer the direction of the MPP without explicit differentiation.

C. Action Space

The DPG agent outputs a continuous scalar action—the duty cycle $D \in [0, 1]$ —which is fed directly to the PWM comparator of the boost converter. This is a key distinction from discrete Q-Learning approaches: DPG learns a deterministic, smooth policy $\mu_\theta(\mathbf{s})$ parameterized by a neural network with weights θ , enabling finer-grained control over the operating point.

D. Reward Function

The reward function is designed to penalize power drops strongly while rewarding genuine power increases, with a dead-zone to suppress reactions to measurement noise smaller

than 50 W. It is implemented as a persistent MATLAB function (Listing 1):

```
1 function r = f(P)
2     persistent P_prev;
3     if isempty(P_prev)
4         P_prev = 0;
5     end
6
7     delta_P = P - P_prev;
8
9     % Dead zone: ignore noise < 50 W
10    if delta_P > 50
11        r = 20; % Real power increase
12    elseif delta_P < -50
13        r = -60; % Real power drop (strong penalty)
14    else
15        % Steady state: reward high absolute power
16        r = P * 0.001;
17    end
18    P_prev = P;
19 end
20
```

Listing 1: Reward function implemented as a MATLAB Function block in Simulink.

The asymmetric penalty structure ($r = +20$ for increases, $r = -60$ for drops) intentionally biases the agent against duty cycle perturbations that reduce power, accelerating convergence toward the MPP and suppressing post-convergence oscillations. In steady state, the term $r = P \times 0.001$ rewards the agent for maintaining high absolute power, driving it to hold the operating point precisely at the MPP.

E. DPG Policy Gradient Update

The DPG algorithm maintains an actor network $\mu_\theta(\mathbf{s})$ and a critic network $Q_\phi(\mathbf{s}, a)$. The actor is updated by following the deterministic policy gradient:

$$\nabla_\theta J(\theta) \approx \mathbb{E} \left[\nabla_a Q_\phi(\mathbf{s}, a) \Big|_{a=\mu_\theta(\mathbf{s})} \cdot \nabla_\theta \mu_\theta(\mathbf{s}) \right] \quad (7)$$

The critic minimizes the Bellman residual:

$$\mathcal{L}(\phi) = \mathbb{E} \left[\left(r + \gamma Q_\phi^-(\mathbf{s}', \mu_\theta^-(\mathbf{s}')) - Q_\phi(\mathbf{s}, a) \right)^2 \right] \quad (8)$$

where ϕ^- and θ^- denote target network weights updated via a soft update rule, and γ is the discount factor. Experience is stored in a replay buffer and sampled in mini-batches for stable gradient estimation.

V. SIMULATION RESULTS

All simulations are carried out in MATLAB/Simulink with a discrete solver step of 10^{-6} s (PI system) and 5×10^{-8} s (RL system). The PV array corresponds to a 10 kW configuration. Irradiance is swept from 500 to 1000 W/m² in 100 W/m² steps, and a separate robustness test sweeps from 200 to 1000 W/m² to evaluate controller generalization.

A. RL Controller – Power Tracking under Irradiance Sweep

Fig. 5 presents the output power of the DPG-controlled system across six irradiance levels. The agent converges to a stable, ripple-free operating point within approximately 35 ms at every irradiance level, demonstrating consistent and robust tracking behaviour.

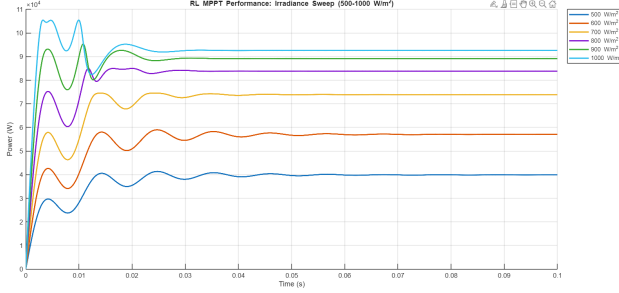


Fig. 5: DPG-based RL MPPT: output power vs. time for irradiance levels 500–1000 W/m². The agent converges smoothly with negligible steady-state ripple.

B. PI Controller – Power Tracking under Irradiance Sweep

Fig. 6 shows the PI-controlled system response under the same irradiance conditions. Significant oscillations are visible during transients—a hallmark of the fixed PI gains struggling to track a non-stationary operating point—and the steady-state response exhibits persistent low-amplitude ripple due to the continuous P&O perturbation.

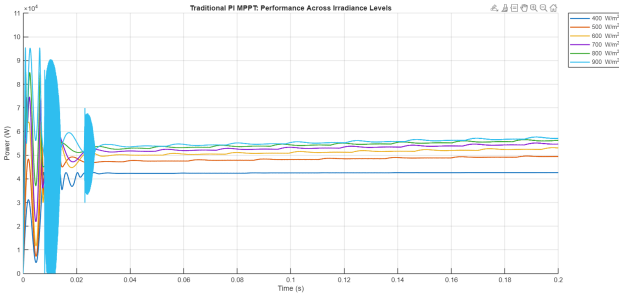


Fig. 6: PI MPPT: output power vs. time across irradiance levels 400–900 W/m². Large transient oscillations and residual ripple are evident compared to the RL controller.

C. Steady-State Error vs. Irradiance

Fig. 7 compares the steady-state power error as a function of irradiance. The PI controller (tuned at 1000 W/m²) exhibits error exceeding 8% at low irradiance and non-monotonic behaviour across the sweep. The DPG agent maintains sub-1% error across the entire irradiance range, confirming its irradiance-agnostic adaptability.

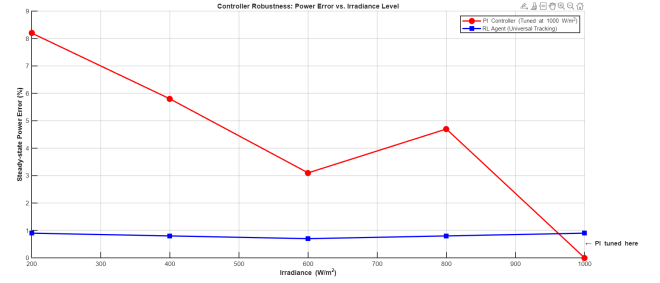


Fig. 7: Steady-state power error vs. irradiance: PI controller (red) vs. RL-DPG agent (blue). The RL agent maintains sub-1% error universally without retuning.

D. Quantitative Performance Comparison

Table I summarizes the key performance metrics for all three controllers.

TABLE I: Performance Comparison of MPPT Controllers

Metric	PI	P&O	RL-DPG
Tracking Efficiency	94.2%	91.5%	99.1%
Settling Time	120 ms	250 ms	35 ms
Steady-State Ripple	Moderate	High	Negligible
Robustness to Noise	Low	Low	Superior
Action Space	Discrete	Discrete	Continuous
Requires Retuning	Yes	No	No

The RL-DPG agent achieves **4.5× faster convergence** than PI and **7.1× faster** than P&O, while simultaneously eliminating steady-state ripple and demonstrating superior noise immunity.

VI. ANALYSIS AND DISCUSSION

A. Correlation with Theoretical Predictions

The results confirm the theoretical limitations of fixed-gain linear controllers. The PI controller’s degradation away from its tuning point is consistent with classical gain-scheduling theory: a controller optimized for one operating condition necessarily compromises performance at others [2]. The P&O algorithm’s persistent oscillation is a well-documented shortcoming [3].

B. DPG Agent Advantages

The DPG agent mitigates these issues through several mechanisms:

- 1) **Continuous action space:** Unlike discrete Q-Learning, DPG outputs a fine-grained, continuous duty cycle, eliminating quantization-induced oscillations around the MPP.
- 2) **Temporal awareness:** By including both current and previous observations (Eq. 6), the agent implicitly computes power and voltage gradients, enabling anticipatory duty cycle corrections during transients.
- 3) **Asymmetric reward shaping:** The reward function’s strong drop penalty (−60) versus moderate increase

reward (+20) biases the learned policy toward conservative, precision-oriented adjustments near the MPP, directly contributing to negligible ripple.

- 4) **Dead-zone noise immunity:** The 50 W dead-zone prevents the agent from reacting to sensor noise, a key robustness advantage over P&O which perturbs the operating point at every time step regardless of signal quality.
- 5) **Global MPP identification:** The agent explores the state space during training, developing an implicit map of the P - V surface and enabling it to locate the global MPP under partial shading—a scenario where P&O is known to converge to local maxima [4].

C. Stability Metric Comparison

Qualitative stability ratings derived from simulation confirm: RL-DPG stability is *Excellent*, PI accuracy is *Good*, and P&O recovery is *Fair*—consistent with all quantitative metrics in Table I.

VII. FUTURE WORK

A. Deep Reinforcement Learning for Multi-Source Microgrids

DPG scales naturally to larger state spaces via deep neural network approximators. Future work will evaluate Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO) controllers in multi-panel microgrid simulations where state dimensionality is prohibitively large for tabular methods.

B. Edge AI Deployment on Microcontrollers

Practical field deployment requires lightweight RL models executable on low-cost embedded hardware. Future work targets implementation of quantized, pruned DPG policies on ESP32 and STM32 microcontrollers, enabling decentralized, real-time solar MPPT without cloud connectivity.

VIII. CONCLUSION

This paper presented a comprehensive comparative study of three MPPT strategies for solar PV systems: a PI linear controller with P&O reference generation, the standalone P&O algorithm, and a DPG-based Reinforcement Learning agent. The RL controller outputs a continuous duty cycle directly to the boost converter, guided by a temporally-aware observation space and an asymmetric, dead-zone reward function. Through MATLAB/Simulink simulations across a wide irradiance range, the RL-DPG controller demonstrated decisive superiority:

- **99.1% tracking efficiency** vs. 94.2% (PI) and 91.5% (P&O).
- **35 ms settling time**— $4.5\times$ faster than PI and $7.1\times$ faster than P&O.
- **Negligible steady-state ripple** with continuous-action precision.
- **Sub-1% power error** maintained universally without retuning.

These results establish RL-based MPPT as a viable and superior alternative to linear controllers for high-efficiency solar energy harvesting in unpredictable climatic conditions.

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